

University Semester Examination Solutions: Electrical Machines-II

(Part-II: 3-Phase Transformers & Tap Changers)

Course Instructor Reference: SR Sir
Electrical Engineering Department

2024

PART-II (Full Marks: 50)

Instructions: Answer any THREE questions including all sub-parts. Question No. 3 carries 18 marks.

Question 1. Correct and/or justify the following:

[4 × 4 = 16 Marks]

Problem Statement 1(i):

Delta-star is more preferable than star-star without neutral three-phase transformer.

Justification:

Thought Process:

1. Recall the primary limitation of a $Y - Y$ connection without a neutral wire: the path for third-harmonic (3^{rd} , 9^{th} , etc.) components of magnetizing current is blocked because $i_{3a} + i_{3b} + i_{3c} = 3I_{3m} \sin(3\omega t) \neq 0$.
2. This suppression of third-harmonic currents forces the core flux to become highly non-sinusoidal (flat-topped).
3. A flat-topped flux containing a large 3^{rd} harmonic component induces a highly peaked sinusoidal voltage in the phase windings, escalating dielectric insulation stress.
4. Conversely, a $\Delta - Y$ configuration provides a closed path within the primary delta mesh loop for third-harmonic currents to circulate freely. Consequently, the core flux remains near-sinusoidal, preventing voltage distortion.

In a three-phase system, the third-harmonic magnetizing current components are co-phasor (in-phase). In an ungrounded star-star ($Y - Y$) transformer, these currents cannot flow because there is no return path ($i_a + i_b + i_c = 0$). As a result, the magnetizing current remains purely sinusoidal, which forces the core magnetic flux $\phi(t)$ to become flat-topped due to magnetic saturation:

$$\phi(t) = \Phi_{1m} \sin(\omega t) - \Phi_{3m} \sin(3\omega t)$$

According to Faraday's law, the induced phase voltage is proportional to $d\phi/dt$:

$$e(t) = -\frac{d\phi}{dt} = -E_{1m} \cos(\omega t) + 3E_{3m} \cos(3\omega t)$$

The factor of 3 scaling the third-harmonic term produces a highly peaked phase-voltage waveform, which can break down insulation.

By contrast, in a Delta-Star ($\Delta - Y$) transformer, the primary delta loop forms a closed mesh circuit where co-phasor third-harmonic currents can freely circulate:

$$I_{\Delta,3rd} = \frac{E_{3rd}}{3 \cdot Z_{3rd}}$$

This circulation supplies the necessary non-linear components to keep the core flux sinusoidal, effectively eliminating the peak voltage stresses on the phase insulation. Hence, $\Delta - Y$ is significantly more stable and widely preferred.

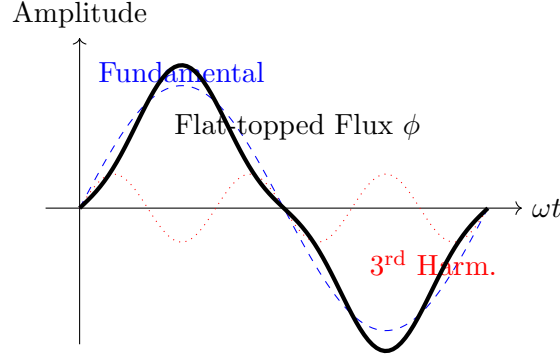


Figure 1: Flux waveform distortion caused by omitting third-harmonic currents in a $Y - Y$ connection.

Problem Statement 1(ii):

The cross section of the central limb of a shell-type transformer is usually designed to have larger cross section to avoid flux saturation.

Justification:

Thought Process:

1. Analyze the core structure of a standard single-phase shell-type transformer.
2. The core consists of a central limb wrapped by the windings and two outer parallel side limbs.
3. The total core flux Φ is generated in the central limb. At the outer joints, it splits equally into two components: $\Phi/2$ flows through the left limb, and $\Phi/2$ flows through the right limb.
4. To maintain a uniform maximum flux density ($B_{\max} = \Phi/A$) across all parts of the magnetic circuit and avoid localized iron saturation, the central limb cross-sectional area must be exactly twice ($2\times$) the area of an individual side limb.

This statement is ****correct**** based on fundamental magnetic circuit balancing. In a shell-type core design, the primary and secondary coils are wound around the central limb. The total magnetic flux $\Phi(t)$ developed inside this central core leg splits symmetrically at the upper and lower yokes into two equal magnetic return paths:

$$\Phi_{\text{left}} = \frac{\Phi}{2}, \quad \Phi_{\text{right}} = \frac{\Phi}{2}$$

To establish an equal maximum operational flux density $B_{\max}$ throughout the frame and optimize material usage without causing magnetic saturation, the cross-sectional areas must satisfy:

$$B_{\max} = \frac{\Phi}{A_{\text{center}}} = \frac{\Phi/2}{A_{\text{outer}}} \implies A_{\text{center}} = 2 \cdot A_{\text{outer}}$$

Thus, designing the central limb with twice the cross-sectional area of the outer limbs prevents excessive core loss and localized magnetic saturation.

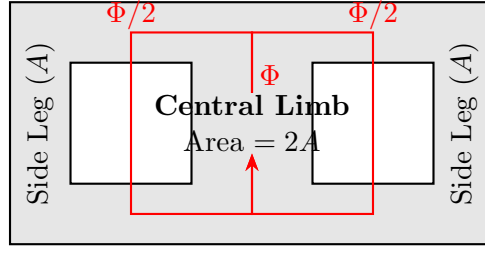


Figure 2: Magnetic flux distribution and cross-sectional area assignment in a shell-type core structure.

Problem Statement 1(iii):

The $9n^{\text{th}}$ harmonics of magnetic flux are always in phase though its $7n^{\text{th}}$ harmonics have a reversed phase sequence compared to fundamental.

Justification:

Thought Process:

1. Let the fundamental balanced three-phase components be defined as:

$$x_a = X_m \sin(\omega t), x_b = X_m \sin(\omega t - 120^\circ), x_c = X_m \sin(\omega t + 120^\circ).$$

2. Calculate the phase displacement for a general k^{th} harmonic component:

$$\theta_b = k \cdot (-120^\circ) \text{ and } \theta_c = k \cdot (+120^\circ).$$

3. For $9n^{\text{th}}$ harmonics ($k = 9n$):

$$\text{Phase B shift} = 9n(-120^\circ) = -1080^\circ n = n \times (-3 \times 360^\circ) \equiv 0^\circ.$$

Thus, $9n^{\text{th}}$ harmonics are in phase across all three lines (Zero Sequence).

4. For $7n^{\text{th}}$ harmonics where $n = 1$ ($k = 7$):

$$\text{Phase B shift} = 7(-120^\circ) = -840^\circ = -120^\circ - 720^\circ \equiv -120^\circ.$$

$$\text{Phase C shift} = 7(+120^\circ) = +840^\circ = +120^\circ + 720^\circ \equiv +120^\circ.$$

This yields an $A - B - C$ sequence, which matches the fundamental positive phase sequence, contradicting the statement's claim of a reversed sequence.

Let us analyze the phase sequences mathematically using the fundamental baseline definitions:

$$\phi_a(t) = \Phi_m \sin(\omega t)$$

$$\phi_b(t) = \Phi_m \sin(\omega t - 120^\circ)$$

$$\phi_c(t) = \Phi_m \sin(\omega t + 120^\circ)$$

For any integer harmonic order k , the relative electrical phase displacement of Phase B and Phase C relative to Phase A is evaluated as:

$$\Delta\theta_b = k \cdot (-120^\circ), \quad \Delta\theta_c = k \cdot (+120^\circ)$$

1. **Case of $9n^{\text{th}}$ order harmonics ($k = 9, 18, 27, \dots$):**

$$\Delta\theta_b = 9n(-120^\circ) = -1080^\circ \cdot n = n \cdot (-3 \times 360^\circ) \equiv 0^\circ \pmod{360^\circ}$$

$$\Delta\theta_c = 9n(+120^\circ) = +1080^\circ \cdot n = n \cdot (+3 \times 360^\circ) \equiv 0^\circ \pmod{360^\circ}$$

Because the phases are offset by integer multiples of 360° , all $9n^{\text{th}}$ harmonic components line up perfectly in phase. They form a ****zero phase sequence****, verifying the first part of the statement.

2. **Case of $7n^{\text{th}}$ order harmonics (for $n = 1, k = 7$):**

$$\Delta\theta_b = 7(-120^\circ) = -840^\circ = -120^\circ - (2 \times 360^\circ) \equiv -120^\circ$$

$$\Delta\theta_c = 7(+120^\circ) = +840^\circ = +120^\circ + (2 \times 360^\circ) \equiv +120^\circ$$

This produces a sequence layout of $A \rightarrow B \rightarrow C$, which is the ****positive phase sequence**** (matching the fundamental).

Therefore, the statement is **partially incorrect**: while the $9n^{\text{th}}$ harmonics are co-phasor (in phase), the 7^{th} harmonic component preserves the fundamental phase sequence rather than reversing it (it is the 5^{th} and 11^{th} harmonics that exhibit a reversed, negative sequence).

Problem Statement 1(iv):

Vector group formation is unavoidable with 30° phase shift in case of Delta-star or Star-delta transformer.

Justification:

Thought Process:

1. Examine the geometric phase relationship between line and phase values in a Delta (Δ) network versus a Star (Y) network.
2. In a Star winding, the line-to-neutral phase voltage $\vec{V}_{AN}$ and the line-to-line voltage $\vec{V}_{AB} = \vec{V}_{AN} - \vec{V}_{BN}$ are structurally offset by exactly 30° in time phase.
3. In a Delta winding, line voltage matches the phase winding voltage directly ($\vec{V}_{AB} = \vec{V}_{ph}$).
4. When linking a primary Δ to a secondary Y, the secondary line outputs are generated from vector differences of phase windings that are locked in phase with the primary delta lines. This structural link introduces an unavoidable geometric phase shift of $\pm 30^\circ$ (corresponding to index marks 1 or 11 on a clock face).

This assertion is **correct**. In any three-phase winding transformation, the phase voltage developed on the secondary side must remain perfectly in phase with the corresponding driving phase voltage on the primary side, since they share the same magnetic core limb.

- In a **Delta (Δ)** winding configuration: The output line-to-line voltage vector corresponds directly to the phase winding voltage vector: $\vec{V}_{\text{line},\Delta} = \vec{V}_{\text{phase},\Delta}$.
- In a **Star (Y)** winding configuration: The output line-to-line voltage vector is derived as the geometric difference between two adjacent phase-to-neutral vectors: $\vec{V}_{AB} = \vec{V}_{AN} - \vec{V}_{BN}$.

From the phasor geometry of a balanced star connection, this vector subtraction creates a fixed angular advancement or delay of exactly 30° :

$$\vec{V}_{AB} = \sqrt{3}V_{AN} \angle +30^\circ$$

Because one side of the transformer is delta-connected and the other is star-connected, this internal 30° offset cannot be cancelled out. This creates an inherent line-to-line voltage phase shift between primary and secondary, leading to the designation of asymmetrical vector groups such as **Dy1** (-30° shift) or **Dy11** ($+30^\circ$ shift).

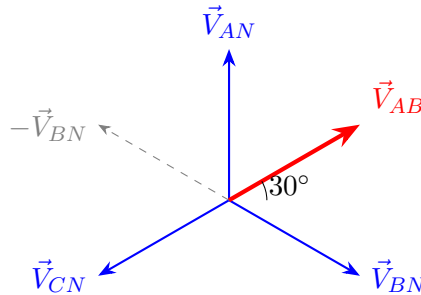


Figure 3: Phasor geometry inside a Star (Y) winding producing a fixed 30° line-to-phase displacement.

Question 2. Core Construction & Vector Groups

[4 + 9 + 3 = 16 Marks]

Problem Statement 2(i):

How core construction helps in reducing harmonics in a three phase transformer?

Answer:

Thought Process:

1. Differentiate between a 3-phase bank of three independent single-phase units (independent magnetic circuits) and an integrated 3-limb core frame (dependent magnetic circuit).
2. In a 3-limb core structure, the three phases share common return legs.
3. Third-harmonic fluxes $\phi_{3a}, \phi_{3b}, \phi_{3c}$ are completely in phase with each other. Thus, at any instant, they attempt to flow in the same direction through all three vertical core limbs simultaneously.
4. Because there is no fourth iron leg to serve as a return path, these fluxes are forced out of the high-permeability iron core into the surrounding high-reluctance paths (transformer oil, insulating medium, and the steel tank walls).
5. The high reluctance ($R_g = l/(\mu_0 A)$) significantly dampens the magnitude of the third-harmonic flux component, making the resultant phase flux much closer to a pure sinusoid.

The specific core architecture of a three-phase transformer plays a critical role in mitigating third-harmonic flux distortion when the winding configuration (such as an ungrounded $Y - Y$ connection) blocks third-harmonic currents.

Consider a standard integrated **3-limb core structure**. The primary fluxes for the three phases are generated within the three parallel vertical core legs. As proven in Question 1(iii), the third-harmonic flux components ($\phi_{3a}, \phi_{3b}, \phi_{3c}$) are co-phasor (fully in phase):

$$\phi_{3a} = \phi_{3b} = \phi_{3c} = \Phi_{3m} \sin(3\omega t)$$

Consequently, at any given instant, the third-harmonic fluxes in all three limbs point in the same direction (e.g., all flowing vertically upward). According to Kirchhoff's Flux Law, these fluxes must find a return path to form a closed magnetic loop.

However, in a 3-limb core, there is no separate common iron leg to carry this return flux. As a result, the third-harmonic fluxes are forced to exit the high-permeability steel core and return through the surrounding high-reluctance pathways, including the transformer oil, structural insulation, and the steel outer tank walls.

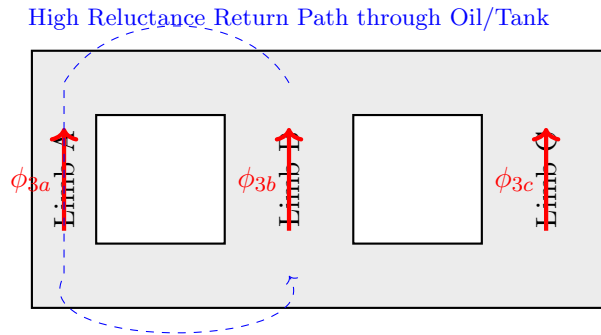


Figure 4: Co-phasor third-harmonic fluxes forced into high-reluctance paths in a 3-limb core.

The reluctance R_{path} of the oil and air gap is several orders of magnitude higher than that of the silicon steel core (R_{core}):

$$R_{\text{path}} = \frac{l}{\mu_0 A} \gg R_{\text{core}} = \frac{l}{\mu_0 \mu_r A}$$

Because the magnetic path has high reluctance, the magnitude of the third-harmonic flux component is severely attenuated:

$$\Phi_{3m} = \frac{\mathcal{F}_{3m}}{R_{\text{path}}} \approx 0$$

By suppressing the third-harmonic flux component, the total core flux remains near-sinusoidal. This protects the phase induced voltages from highly peaked distortions, effectively eliminating harmonic resonance risks without requiring an external neutral wire.

Problem Statement 2(ii):

Draw the connection diagram and phasor diagram of the following connections:

a) Dy1, b) Yz1 and c) Yd1

Answer:

Thought Process:

1. Decode each vector group using standard clock-face nomenclature:
 - **Dy1**: Primary is Delta (Δ), Secondary is Star (Y). The secondary line voltage lags the primary line voltage by 30° (1 o'clock position $\Rightarrow -30^\circ$ shift).
 - **Yz1**: Primary is Star (Y), Secondary is Zigzag (z). The secondary output voltage lags by 30° (1 o'clock position).
 - **Yd1**: Primary is Star (Y), Secondary is Delta (Δ). The secondary line voltage lags by 30° (1 o'clock position).
2. For each group, construct:
 - The primary and secondary winding layout, showing explicit terminal labels (A_1, A_2 or a_1, a_2) and connections.
 - The corresponding geometric phasor diagrams aligned to match the 30° lagging requirement.

The clock convention specifies that the primary line voltage phasor is anchored as the reference pointer pointing directly at 12 o'clock (0°). An index rating of **1** implies that the secondary line voltage phasor is oriented at 1 o'clock, corresponding to an angular lag of exactly 30° behind the primary reference.

a) Vector Group Dy1

- **Primary Winding (Δ)**: Phases connected in a closed delta ring: $A_2 \rightarrow B_1, B_2 \rightarrow C_1, C_2 \rightarrow A_1$.
- **Secondary Winding (Y)**: Terminals a_1, b_1, c_1 are shorted together to form the isolated common neutral point (n). The output lines are fed directly from terminals a_2, b_2, c_2 .



Figure 5: Dy1 Winding Connection Diagram.

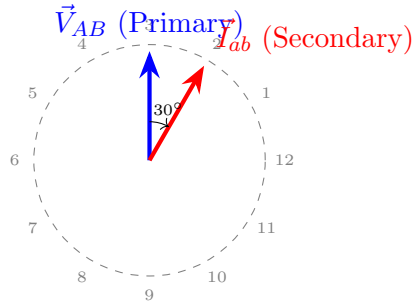


Figure 6: Dy1 Phasor Clock Face (Secondary line lagging by 30°).

b) Vector Group Yz1

In a Zigzag (z) connection, each secondary phase winding is split into two equal halves distributed across different core legs, connected in series-opposition. This cancels out zero-sequence third-harmonic components.

- **Primary Winding (Y):** A_1, B_1, C_1 are linked to form the primary neutral point N .
- **Secondary Winding (z):** Winding segments are cross-connected to ensure the resultant phase vector shifts to lag by -30° .

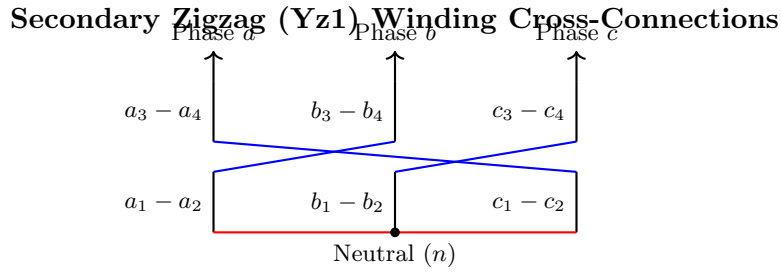


Figure 7: Winding layout and cross-connections for a Yz1 vector group.

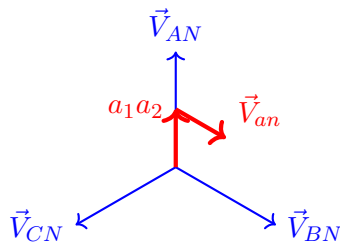


Figure 8: Phasor derivation of the Yz1 configuration showing the 30° lagging output.

c) Vector Group Yd1

- **Primary Winding (Y):** Symmetrical star with a central isolated neutral point (N).
- **Secondary Winding (Δ):** Windings are connected in a closed loop ($a_2 \rightarrow b_1, b_2 \rightarrow c_1, c_2 \rightarrow a_1$) configured to shift the output line voltages by -30° .



Figure 9: Yd1 Connection Layout Diagram.

Problem Statement 2(iii):

Yy0 and Dd6 connected transformers cannot be connected in parallel- explain.

Answer:

Thought Process:

1. Recall the primary criteria for paralleling transformers: same voltage ratio, identical polarities, same phase sequence, and zero relative phase displacement between secondary line terminals.
2. Analyze the specific phase displacement of the two given vector configurations:
 - **Yy0**: Line-to-line phase displacement is exactly 0° (12 o'clock).
 - **Dd6**: Line-to-line phase displacement is exactly 180° (6 o'clock).
3. Determine the phase angle difference between the two configurations: $\Delta\theta = 180^\circ - 0^\circ = 180^\circ$.
4. Evaluate the resulting secondary line-to-line potential difference ($\Delta\vec{V}$). Since the voltage vectors are in direct opposition, $\Delta\vec{V} = \vec{V}_{2,Yy0} - \vec{V}_{2,Dd6} = \vec{V}_2 - (-\vec{V}_2) = 2\vec{V}_2$.
5. A potential difference of $2\vec{V}_2$ across the parallel secondary connection acts as a direct line-to-line short circuit, causing catastrophic overcurrent damage.

To safely operate two three-phase transformer banks in parallel, the secondary line-to-line output voltages must have zero relative phase displacement at the parallel busbars ($\Delta\theta = 0^\circ$).

1. A **Yy0** transformer exhibits a phase shift index of 0, meaning the secondary line voltages are perfectly in phase with the primary references:

$$\vec{V}_{ab,Yy0} = V_2 \angle 0^\circ$$

2. A **Dd6** transformer exhibits a phase shift index of 6, meaning the secondary line voltages are inverted by 180° relative to the primary references:

$$\vec{V}_{ab,Dd6} = V_2 \angle 180^\circ = -V_2 \angle 0^\circ$$

If these two units are paralleled on the same busbars, the net voltage difference $\Delta\vec{V}$ driving current through the local loop is the sum of their magnitudes:

$$\Delta\vec{V} = \vec{V}_{ab,Yy0} - \vec{V}_{ab,Dd6} = V_2 \angle 0^\circ - (-V_2 \angle 0^\circ) = 2V_2 \angle 0^\circ$$

$$\begin{array}{c} \vec{V}_{ab,Dd6} = V_2 \angle 180^\circ \quad \leftarrow \text{Red arrow pointing left} \\ \vec{V}_{ab,Yy0} = V_2 \angle 0^\circ \quad \rightarrow \text{Blue arrow pointing right} \\ \hline \Delta\vec{V} = 2V_2 \text{ (Direct Line-to-Line Short Circuit)} \end{array}$$

Figure 10: Voltage vector opposition when attempting to parallel Yy0 and Dd6 transformers.

Because the leakage impedance of a transformer is low, this $2V_2$ potential difference will drive a massive short-circuit loop current through the windings:

$$I_{\text{short}} = \frac{2V_2}{Z_{e1} + Z_{e2}}$$

This severe overcurrent will cause catastrophic thermal and mechanical breakdown of both transformer winding assemblies. Consequently, parallel operation of these two configurations is physically impossible.

Question 3. Scott Connection

[7 + 11 = 18 Marks]

Problem Statement 3(i):

Show that by using two single phase transformers with proper taps, three phase power supply is possible from a two phase supply.

Answer:

Thought Process:

1. Identify the schematic layout of a Scott T-connection network.
2. The system requires two distinct single-phase transformer units:
 - The ****Main Transformer****, connected directly across lines B and C , with a center tap N at 50% of its primary winding.
 - The ****Teaser Transformer****, connected between phase terminal A and the center tap point N of the main transformer primary.
3. Use phasor geometry to derive the required turn tap ratios:
 - Let the balanced 3-phase primary line-to-line inputs be $\vec{V}_{AB}, \vec{V}_{BC}, \vec{V}_{CA}$.
 - The main primary voltage is $\vec{V}_{BC} = V_L \angle 0^\circ$.
 - The teaser primary voltage operates along the perpendicular median axis: $\vec{V}_{AN} = \frac{\sqrt{3}}{2} V_L \angle 90^\circ \approx 0.866 V_L \angle 90^\circ$.
4. To establish perfectly balanced secondary induced voltage magnitudes ($V_{2,\text{main}} = V_{2,\text{teaser}}$), the number of primary turns must be adjusted. The teaser primary must have 86.6% of the total turns of the main primary winding ($N_T = 0.866 N_M$).

The Scott connection uses two single-phase transformers with asymmetrical primary turn ratios—the ****Main**** transformer and the ****Teaser**** transformer—to convert power between three-phase and two-phase configurations.

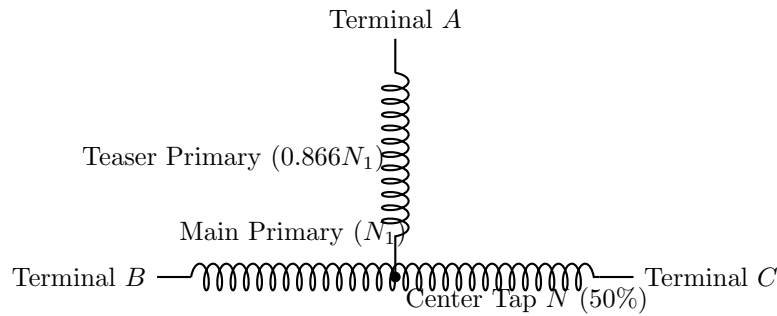


Figure 11: Winding configuration schematic for the primary side of a Scott T-connection.

Let the primary side be connected to a balanced three-phase line supply with line-to-line voltage V_L . Taking the line voltage $\vec{V}_{BC}$ across the Main transformer as the reference baseline:

$$\vec{V}_{BC} = V_L \angle 0^\circ$$

The voltage from line terminal A to the neutral center-tap point N corresponds to the geometric median altitude of an equilateral triangle:

$$\vec{V}_{AN} = \vec{V}_{AB} + \vec{V}_{BN} = V_L \angle 120^\circ + \frac{V_L}{2} \angle 180^\circ$$

$$\vec{V}_{AN} = \left(-\frac{V_L}{2} + j\frac{\sqrt{3}}{2}V_L \right) - \frac{V_L}{2} = -V_L + j\frac{\sqrt{3}}{2}V_L \implies \text{relative to tap } N :$$

$$\vec{V}_{AN} = \frac{\sqrt{3}}{2}V_L \angle 90^\circ \approx 0.866V_L \angle 90^\circ$$

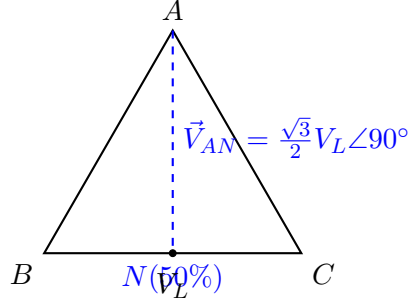


Figure 12: Geometric voltage vector mapping inside the primary triangle of a Scott connection.

Let the secondary winding of both the Main and Teaser transformers be wound with identical turns N_2 , yielding the following secondary induced output voltages:

1. ****Main Secondary Output (V_{2M}):****

$$V_{2M} = \frac{N_2}{N_1} \cdot V_{BC} = \frac{N_2}{N_1} V_L \angle 0^\circ$$

2. ****Teaser Secondary Output (V_{2T}):**** To generate a balanced two-phase output, the teaser secondary voltage magnitude must equal the main secondary voltage magnitude ($|V_{2T}| = |V_{2M}|$). Since its primary voltage is restricted to $0.866V_L$, its primary turns N_T must be reduced to:

$$V_{2T} = \frac{N_2}{N_T} \cdot \left(\frac{\sqrt{3}}{2}V_L \right) \equiv \frac{N_2}{N_1} V_L \implies N_T = \frac{\sqrt{3}}{2}N_1 \approx 0.866N_1$$

Using an 86.6% primary winding tap on the teaser transformer gives two secondary output voltages with equal magnitudes and a 90° phase separation:

$$\vec{V}_{2M} = V_2 \angle 0^\circ, \quad \vec{V}_{2T} = V_2 \angle 90^\circ$$

This establishes a balanced two-phase output system from a three-phase input.

Problem Statement 3(ii):

In Scott-connected transformers, teaser transformer supplies 0.75 leading power factor load of 25 kW at 220 V and main transformer supplies 0.75 power factor lagging load of 45 kW at 220 V, from a three phase input line voltage of 3300V. Determine the input 3-phase line currents. Neglect magnetizing currents and the leakage impedance drops. Draw input current phasors computed for above along with a Scott-connection diagram.

Answer:

Thought Process:

1. Extract the given numerical constraints:

- Primary Line Voltage $V_L = 3300$ V.
- Secondary Voltage $V_2 = 220$ V.
- Main Winding Turn Ratio: $N_1/N_2 = 3300/220 = 15$.
- Teaser Load: $P_T = 25$ kW at $\cos \theta_T = 0.75$ leading $\implies \theta_T = +41.41^\circ$.
- Main Load: $P_M = 45$ kW at $\cos \theta_M = 0.75$ lagging $\implies \theta_M = -41.41^\circ$.

2. Calculate secondary load currents:

- $I_{2T} = \frac{25000}{220 \times 0.75} = 151.52$ A.
- $I_{2M} = \frac{45000}{220 \times 0.75} = 272.73$ A.

3. Express currents as complex vectors relative to the phase references:

- $\vec{I}_{2T} = 151.52 \angle +41.41^\circ$ A (relative to teaser voltage axis).
- $\vec{I}_{2M} = 272.73 \angle -41.41^\circ$ A (relative to main voltage axis).

4. Map the secondary currents back to primary line currents $\vec{I}_A, \vec{I}_B, \vec{I}_C$:

- $\vec{I}_A = \frac{N_2}{0.866N_1} \vec{I}_{2T} = \frac{2}{\sqrt{3} \cdot 15} \cdot 151.52 = 11.66$ A.

- The currents in lines B and C are superpositions of the main load current and the returning teaser current split equally:

$$\vec{I}_B = \frac{N_2}{N_1} \vec{I}_{2M} - \frac{1}{2} \vec{I}_A$$

$$\vec{I}_C = -\frac{N_2}{N_1} \vec{I}_{2M} - \frac{1}{2} \vec{I}_A$$

5. Perform vector calculations to determine the exact magnitudes and phase angles of the primary line currents.

Step 1: Compute secondary current magnitudes and vector expressions

$$\cos \theta = 0.75 \implies \theta = \cos^{-1}(0.75) = 41.41^\circ$$

- **Teaser Secondary Current (I_{2T}):**

$$I_{2T} = \frac{P_T}{V_2 \cdot \cos \theta_T} = \frac{25000}{220 \times 0.75} = 151.52 \text{ A}$$

Since the power factor is leading, the current vector relative to the teaser voltage baseline ($\angle 90^\circ$) is:

$$\vec{I}_{2T} = 151.52 \angle (90^\circ + 41.41^\circ) = 151.52 \angle 131.41^\circ \text{ A}$$

- **Main Secondary Current (I_{2M}):**

$$I_{2M} = \frac{P_M}{V_2 \cdot \cos \theta_M} = \frac{45000}{220 \times 0.75} = 272.73 \text{ A}$$

Since the power factor is lagging, the current vector relative to the main voltage baseline ($\angle 0^\circ$) is:

$$\vec{I}_{2M} = 272.73 \angle (0^\circ - 41.41^\circ) = 272.73 \angle -41.41^\circ \text{ A}$$

Step 2: Calculate the primary line current vectors The baseline turns transformation factor for the main core assembly is:

$$v_{\text{ratio}} = \frac{N_2}{N_1} = \frac{220}{3300} = \frac{1}{15}$$

1. **Line Current $\vec{I}_A$:** Line A feeds the teaser primary directly.

$$\vec{I}_A = \frac{2}{\sqrt{3}} \cdot \frac{N_2}{N_1} \cdot \vec{I}_{2T} = \frac{2}{\sqrt{3} \times 15} \times 151.52 \angle 131.41^\circ = 11.66 \angle 131.41^\circ \text{ A}$$

Converting this to rectangular coordinates:

$$\vec{I}_A = 11.66(\cos(131.41^\circ) + j \sin(131.41^\circ)) = -7.71 + j8.72 \text{ A}$$

2. **Line Current $\vec{I}_B$:** Winding geometry dictates that line current B combines the transformed main current and half of the returning teaser current:

$$\vec{I}_B = \frac{N_2}{N_1} \vec{I}_{2M} - \frac{1}{2} \vec{I}_A$$

$$\frac{N_2}{N_1} \vec{I}_{2M} = \frac{272.73 \angle -41.41^\circ}{15} = 18.18 \angle -41.41^\circ = 13.64 - j12.02 \text{ A}$$

$$-\frac{1}{2} \vec{I}_A = -\frac{1}{2}(-7.71 + j8.72) = 3.86 - j4.36 \text{ A}$$

Combining these terms gives:

$$\vec{I}_B = (13.64 + 3.86) + j(-12.02 - 4.36) = 17.50 - j16.38 \text{ A}$$

Converting back to polar form:

$$\vec{I}_B = \sqrt{17.50^2 + (-16.38)^2} \angle \tan^{-1} \left(\frac{-16.38}{17.50} \right) = \mathbf{23.97 \angle -43.11^\circ \text{ A}}$$

3. **Line Current $\vec{I}_C$:** Line current C is the vector sum of the inverted main current component and the remaining half of the returning teaser current:

$$\vec{I}_C = -\frac{N_2}{N_1} \vec{I}_{2M} - \frac{1}{2} \vec{I}_A$$

$$-\frac{N_2}{N_1} \vec{I}_{2M} = -13.64 + j12.02 \text{ A}$$

$$\vec{I}_C = (-13.64 + 3.86) + j(12.02 - 4.36) = -9.78 + j7.66 \text{ A}$$

Converting back to polar form:

$$\vec{I}_C = \sqrt{(-9.78)^2 + 7.66^2} \angle \left(180^\circ - \tan^{-1} \left(\frac{7.66}{9.78} \right) \right) = \mathbf{12.42 \angle 141.96^\circ \text{ A}}$$

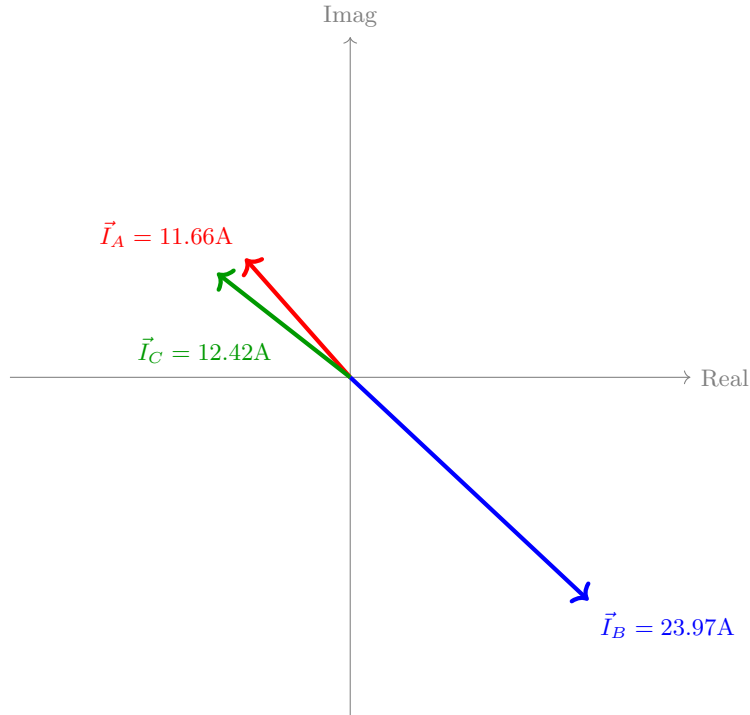


Figure 13: Calculated primary three-phase input line current vectors.

Question 4. Tap Changers

[7 + 5 + 4 = 16 Marks]

Problem Statement 4(i):

Draw and explain two resistor type on-load tap changer schemes.

Answer:

On-load tap changers (OLTC) allow the transformer turns ratio to be adjusted while the transformer is energized and carrying load. This is achieved without interrupting the load current or short-circuiting parts of the winding.

1. Diverter Switch Scheme with Two Resistors (Symmetrical Flag Cycle)

This configuration splits the process into two separate assemblies: a selector switch that selects the next tap under zero-current conditions, and a high-speed diverter switch that handles current interruption and arcing.

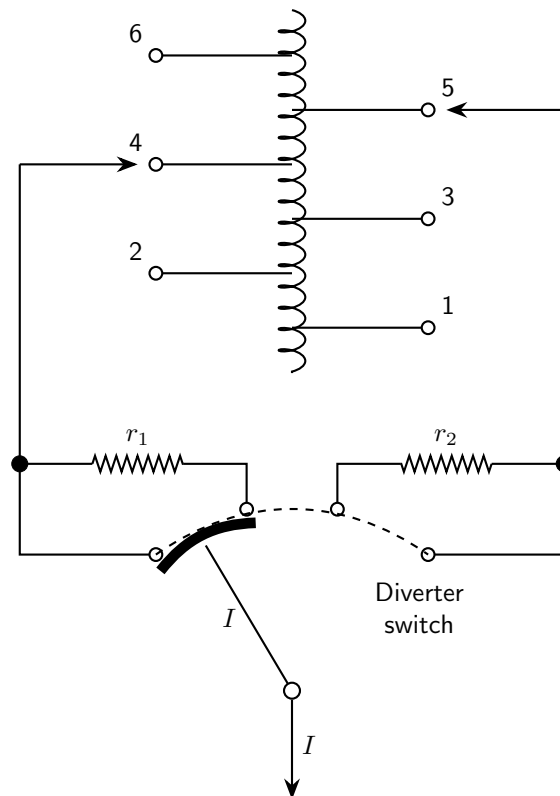


Fig : Diverter Switch

Transition Sequence Steps (From Tap 1 to Tap 2):

1. **Position 1:** The diverter arm rests on main contact M_1 . Load current flows directly from Tap 1 through selector contact A, bypassing the transition resistors.
2. **Position 2:** Selector contact B closes onto Tap 2 under zero current. Winding section 1-2 remains open circuit at contact M_2 .
3. **Position 3:** The high-speed diverter spring mechanism triggers, snapping the arm downward. The contact leaves M_1 and bridges across both transition resistors R_1 and R_2 . The load current divides between both paths, and the short-circuit current between Tap 1 and Tap 2 is limited by $(R_1 + R_2)$.
4. **Position 4:** The arm continues moving downward, opening the Tap 1 path. The load current now flows entirely from Tap 2 through resistor R_2 .

5. **Position 5:** The arm completes its travel and locks onto main contact M_2 . This short-circuits and bypasses resistor R_2 , delivering full load current directly from Tap 2.

2. Selector-Switch Scheme with One Resistor (Asymmetrical Pennant Cycle)

This compact configuration combines the selection and current diversion tasks into a single moving arm mechanism, using only one transition resistor (R).

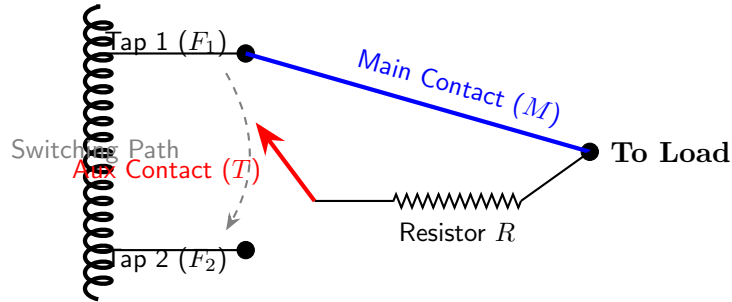


Figure 14: Optimized selector-switch layout with an isolated, single-resistor branch implementing an asymmetrical Pennant Cycle.

Transition Sequence Steps (From Tap 1 to Tap 2):

1. **Position 1:** Main contact M rests on fixed tap contact F_1 . Load current flows directly to the output busbar. The auxiliary transition contact T is open.
2. **Position 2:** The auxiliary contact T moves forward and closes onto fixed contact F_2 . Winding section 1-2 is now bridged by the single resistor R , which limits the circulating current.
3. **Position 3:** The main contact M snaps open, breaking its connection with F_1 . The load current is temporarily diverted through the single resistor R and auxiliary contact T from Tap 2.
4. **Position 4:** The main contact M continues its rotation and closes onto fixed contact F_2 . This safely short-circuits and bypasses the resistor branch. Auxiliary contact T then opens, completing the cycle.

Problem Statement 4(ii):

Write the advantages and disadvantages of using off-load and on-load type tap-changers.

Answer:

Table 1: Comparative Analysis of Off-Load vs. On-Load Tap Changing Systems.

Type	Advantages	Disadvantages
Off-Load Tap Changer	• Highly compact mechanical design. • Lower initial capital cost. • Eliminates transition resistors and arcing control systems.	• Requires isolating the transformer from the grid, causing power interruptions. • Cannot perform real-time voltage regulation under varying load conditions.
On-Load Tap Changer	• Enables real-time voltage regulation under load without service interruptions. • Supports automated grid voltage optimization.	• Mechanically complex with higher capital costs. • Regular contact arcing requires frequent oil filtering and maintenance.

Problem Statement 4(iii):

Why high voltage winding of a transformer is preferred for tappings of tap-changer?

Answer:

Thought Process:

1. Examine the electrical properties of High-Voltage (HV) versus Low-Voltage (LV) windings:
 - $V_{HV} \gg V_{LV} \implies I_{HV} \ll I_{LV}$ (for a constant power rating $S = \sqrt{3}VI$).
2. Evaluate the mechanical impacts of lower current ratings on the tap changer contacts:
 - Smaller current reduces the thermal dissipation (I^2R) and arcing intensity across the contacts during tap transitions, which extends contact operational life.
3. Consider structural winding accessibility:
 - In standard concentric transformer designs, the LV winding is placed closest to the core limb to reduce insulation requirements. The HV winding is wound over the outside of the LV winding, making its turns structurally accessible for physical tap terminations.
4. Assess turn granularity:
 - The HV winding has a larger total number of turns ($N_{HV} \gg N_{LV}$). This provides finer, more precise voltage control per step tap modification compared to the coarse adjustments available on the LV side.

The high-voltage (HV) winding assembly is preferred for tap changer installations due to several engineering design factors:

1. **Arcing and Thermal Dissipation Benefits:** Because the operating voltage is high, the nominal line current on the HV side is significantly lower than on the LV side ($I = S/(\sqrt{3}V)$). This lower current reduces the energy of arcs generated during tap transitions, lowering contact wear and reducing thermal losses (I^2R) across the switch mechanism.
2. **Mechanical Accessibility:** In standard concentric coil arrangements, the LV winding is wound closest to the grounded steel core frame to minimize insulation volume. The HV winding is positioned as the outermost layer, making it physically accessible for running mechanical tap leads to the external tap changer switchbox.

3. **Voltage Regulation Resolution:** The HV winding has a much higher total number of turns ($N_{HV} \gg N_{LV}$). This large number of turns allows for fine adjustments to the turns ratio, providing high resolution voltage regulation:

$$\Delta V_{\text{step}} = \frac{1}{N_{HV}} \cdot 100\%$$

Question 5. Short Notes

[8 × 2 = 16 Marks]

Problem Statement 5(i):

Development of voltage stress along the winding of a three phase transformer for input impulse and RMS voltage. And also write the measure to be taken to withstand it.

Answer:

Thought Process:

1. Analyze how a standard power frequency RMS voltage (50 Hz) distributes across a transformer winding. At low frequencies, inductive impedance dominates ($\omega L \gg 1/(\omega C)$), resulting in a linear, uniform voltage gradient from the line end to the grounded neutral terminal.
2. Analyze the response to a high-frequency high-voltage lightning surge or impulse wave (steep wavefront with equivalent frequency $f \approx 1$ MHz).
3. At high frequencies, the series inductance acts as an open circuit, and the behavior is dominated by the winding's capacitance network. This includes the series capacitance between adjacent coil turns (C_s) and the shunt capacitance from the turns to the grounded core/tank frame (C_g).
4. Set up the differential equation for the capacitance ladder network to find the initial voltage distribution: $V(x) = V_0 \frac{\sinh(\alpha x)}{\sinh(\alpha l)}$, where $\alpha = \sqrt{C_g/C_s}$.
5. This creates a non-uniform exponential voltage gradient. The turns closest to the line end experience high dielectric voltage stresses.
6. Identify standard mitigation methods: electrostatic shielding plates, interleaved winding styles, and line end surge arrestors.

1. Power Frequency RMS Voltage Distribution (50 Hz): Under steady-state power frequency conditions, the internal voltage distribution along the transformer winding is governed by the winding's series inductance. The inductive reactance dominates over the internal capacitive networks ($\omega L \gg 1/(\omega C)$). As a result, the voltage drops uniformly along the length of the winding, creating a constant voltage gradient ($dV/dx = \text{constant}$).

2. High Voltage Surge Impulse Distribution (Lightning Transients): A lightning impulse wave features a steep wavefront that rises to its peak in approximately $1.2 \mu\text{s}$, which corresponds to an equivalent high-frequency transient ($f \approx 1$ MHz). At these frequencies, the inductive paths act as open circuits ($X_L = 2\pi f L \rightarrow \infty$).

The initial voltage distribution is therefore determined by the transformer's internal capacitance network, which consists of the series capacitance between adjacent winding turns (C_s) and the shunt capacitance between the turns and the grounded steel core or tank structure (C_g).

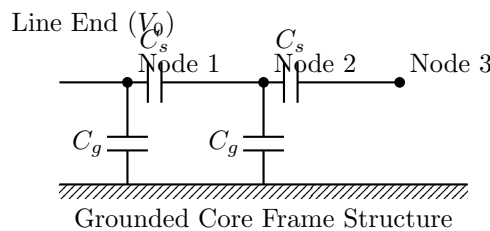


Figure 15: Capacitive ladder network model of a transformer winding under high-frequency transient stresses.

The shunt capacitances (C_g) draw charging currents to ground, causing the current through the series capacitances (C_s) to decrease along the winding length. This non-uniform current produces an exponential voltage distribution described by the following equation:

$$V(x) = V_0 \frac{\sinh(\alpha \frac{x}{l})}{\sinh(\alpha)}$$

$$\text{where } \alpha = \sqrt{\frac{C_g}{C_s}}$$

The parameter α typically ranges between 5 and 15 in standard power transformers. This large value concentrates the initial voltage drop across the first few turns near the line end of the winding, subjecting them to severe dielectric stress.

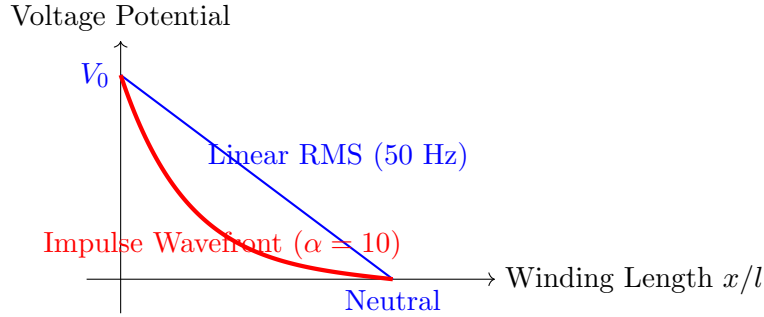


Figure 16: Voltage distribution profiles: linear drop at power frequency (RMS) vs. exponential concentration under an impulse surge.

Mitigation Measures to Protect the Windings:

1. **Interleaved Winding Design:** Interleaving the winding conductors alters the physical arrangement of the turns to place non-consecutive turns adjacent to one another. This increases the effective series capacitance C_s , which reduces the value of $\alpha = \sqrt{C_g/C_s}$ and flattens the initial impulse distribution curve.
2. **Electrostatic Shielding (Static Ring):** Fitting a grounded metal shield ring around the initial line-end sections introduces an auxiliary capacitance that injects compensating currents into the winding nodes. This offsets the current lost through the shunt capacitances (C_g), restoring a near-linear initial voltage distribution.

Problem Statement 5(ii):

Tertiary windings and its applications.

Answer:

Thought Process:

1. Define a tertiary winding: It is a third auxiliary winding wound onto the same core limb alongside the primary and secondary windings, usually connected in a closed Delta (Δ) loop configuration.
2. Identify the main operational problems it solves in ungrounded star-star ($Y-Y$) systems:
 - It provides a path for circulating co-phasor third-harmonic currents, which restores a sinusoidal core flux and prevents phase voltage distortion.
 - It stabilizes the neutral point position during severely unbalanced three-phase load conditions.
3. List auxiliary application areas: supplying localized substation auxiliary loads, and interfacing with synchronous condensers for reactive power (VAr) compensation.

A tertiary winding is an auxiliary third winding added to a transformer core limb alongside the primary and secondary windings. It is typically connected in a closed Delta (Δ) configuration, creating a three-winding transformer system.

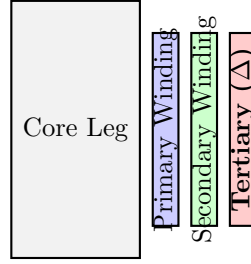


Figure 17: Concentric placement of primary, secondary, and tertiary windings on a single core limb.

Key Engineering Applications and Functions:

1. **Suppression of Third-Harmonic Voltages:** In large star-star ($Y - Y$) power transformer installations, the tertiary winding is connected in a closed delta ring. This closed loop provides a path for co-phaser third-harmonic components of the magnetizing current to circulate freely:

$$I_{\text{tertiary, 3rd}} = \frac{E_{3\text{rd}}}{3 \cdot Z_{\text{winding}}}$$

This circulation supplies the component needed to maintain a sinusoidal core magnetic flux, eliminating the high-frequency voltage peaks that can stress the main insulation.

2. **Neutral Point Stabilization:** If a star-star transformer experiences an unbalanced single line-to-neutral load, the unbalance shifts the magnetic neutral point, distorting the phase voltages. The tertiary delta winding acts as a balancing layer by transferring energy between phases, which counterbalances the zero-sequence currents and stabilizes the neutral point.
3. **Substation Auxiliary Power Supply:** The auxiliary winding can be designed with a reduced voltage rating to supply local substation loads, such as cooling fans, control circuitry, and lighting networks, directly from the main power transformer.
4. **Reactive Power Compensation Interfacing:** High-voltage transmission substations often connect static VAR compensators or synchronous condensers directly across the delta tertiary terminals to inject or absorb reactive power, supporting grid voltage regulation.

— END OF SOLUTIONS —